

Teach-In 2026

by Mike Tooley



World of Wireless – An Introduction to Radio and Wireless Technology, part 12.11

Long-distance radio communications

In Part 10 last month, we introduced short-range wireless devices (SRD) and showed how these versatile chips and modules enable data transmission and reception between remote devices without the need for a wired connection. Our *Hands-On* constructional project featured a simple RF field strength indicator that can be used to make relative measurements up to 450MHz.

This month, we bring our *Teach-In* to a close by providing you with some ideas that will take you further

into the world of radio and wireless technology. Our final *Hands-On* project is a high-gain RF preselector.

Shortwave listening

Shortwave listening is a fascinating pastime enjoyed by many people all over the world, bringing news, documentary features, entertainment and culture to you. Thanks to the ionosphere, you can listen to scheduled broadcasts from distant countries. Shortwave listeners (SWL) tune their sets (almost any HF receiver) to signals in

bands designated for international broadcasting.

Shortwave bands are often described by the wavelength in metres. They include:

- The **49m band (5.9-6.2MHz)** is usually good at night, delivering strong continental stations, as well as those from further afield, including Asia and Central America. It is less active during the day.
- The **41m band (7.2-7.5MHz)** is good during the hours of darkness. It is popular with far-eastern



Fig.11.1: shortwave receivers are widely available.

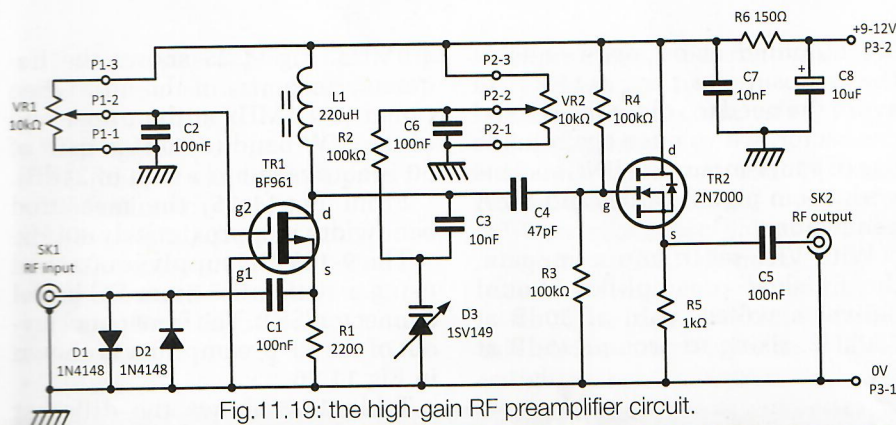
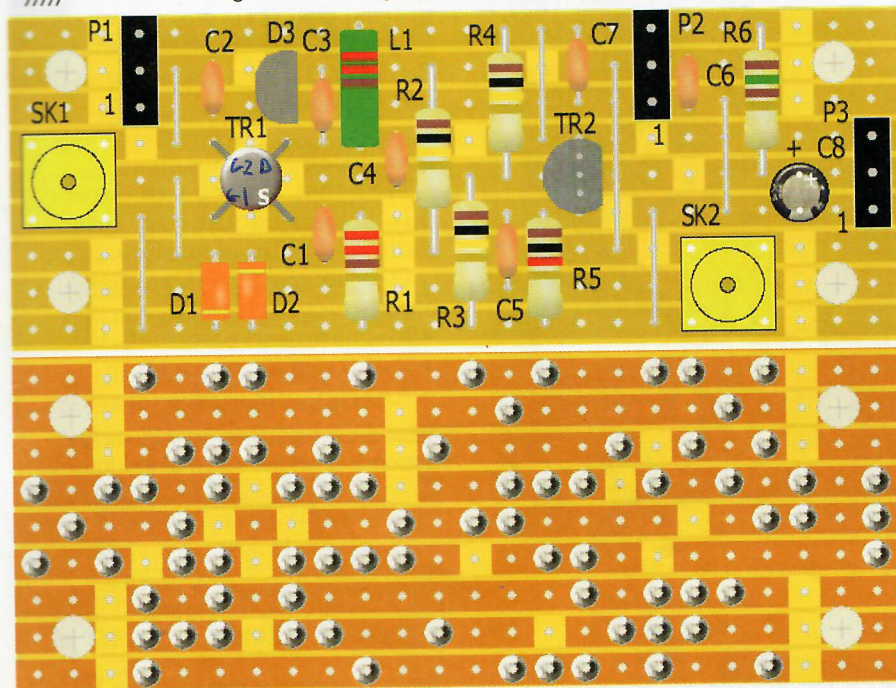


Fig.11.19: the high-gain RF preamplifier circuit.



Figs.11.20 & 21: the high-gain RF preamplifier component layout (top & bottom views).

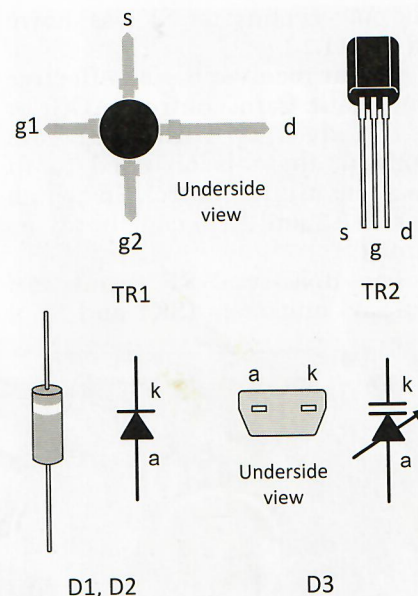


Fig.11.22: the semiconductor pin connections for the high-gain RF preamplifier.

you live anywhere near a busy airport.

2. With a simple quarter-wave vertical antenna or our *Hands-On* dipole reference antenna (Part 5, March 2026), you should have little difficulty locating signals. It's also worth using the internet to check the frequencies being used at your local airport. You should find that several are in regular use.
3. If you plan to try ADS-B reception, it's worth spending a little more to obtain an SDR designed specifically for use with ADS-B. A lower noise figure and front-end filtering can improve ADS-B performance, but results will be disappointing if you intend to use the same dongle for VHF voice and data. If you wish to receive VHF and ADS-B, you may do better purchasing a good quality general-purpose SDR.
4. ADS-B operates at 1.090GHz, so your receiving antenna should be designed specifically for this frequency. Unfortunately, our *Hands-On* dipole reference antenna (described in Part 5 of this

series, in the March 2026 issue) is not suitable for operation at such a high frequency.

Hands-On: A high-gain RF preamp

This month's *Hands-On* project is a high-gain RF preamplifier that's ideal for long-distance (DX) listening on the medium-wave (MW) band. If a different range is required, just a single inductor needs to be changed.

The circuit of the RF preamplifier is shown in Fig.11.19; it comprises a dual-gate Mosfet common-gate amplifier (TR1) followed by a standard Mosfet (TR2) connected as a unity-gain source-follower.

The preamplifier is designed with a low-impedance input suitable for use with a random-length long-wire antenna, with the two anti-parallel silicon switching diodes (D1 & D2) providing over-voltage protection.

A parallel resonant circuit formed by inductor L1 and variable-capacitance diode D3 (in series with DC isolating capacitor C3) forms the load for TR1. The very high impedance of this arrangement, together

with negligible loading from TR2, ensures a very high degree of selectivity.

Source-follower TR2 is suitable for driving a low-impedance output and/or a relatively long output cable run.

Sensitivity is varied by means of a front panel control, VR1, while a second front panel control, VR2, acts as a tuning control.

The preamplifier requires a supply current of less than 10mA from an external 9-12V DC source, such as a plugpack.

As with most of our *Hands-On* projects, it is built on a small piece of perforated copper stripboard measuring 25 × 64 mm, arranged as nine strips each of 24 rows. The component layout (viewed from above) is shown in Fig.11.20, while the corresponding track layout (viewed from below) is in Fig.11.21.

The required 27 track breaks can be made using a spot-face cutter or a small drill bit, while the seven links on the upper side of the board can be made using short lengths of tinned copper wire.

Take care to observe the correct polarities of TR1, TR2 and D2 (see Fig.11.22).

The RF preamplifier can be mounted in a small ABS enclosure. Suitable cases are available from various online suppliers at reasonable prices.

The enclosure used for the prototype measured 100 × 68 × 50mm. The finished stripboard is mounted on the underside of the enclosure lid, together with VR1, VR2, S1, and

the power connector, SK3, as shown in Fig.11.23.

If your receiver has an effective automatic gain control (AGC) or is already fitted with an RF gain control, there is no need to fit the sensitivity control, in which case P1-2 and P1-3 can simply be linked.

The BNC-type RF input and output connectors (SK1 and SK2)

are mounted at opposite ends of the enclosure, as per Fig.11.24. To avoid the need to solder the coaxial connector, we've once again made use of panel-mounting BNC sockets with 15cm pigtail cables and SMA connectors.

With VR1 set to minimum gain, the finished preamplifier should deliver a voltage gain of 30dB at 750kHz, rising to around 35dB at

1.9MHz. Fig.11.25 shows the frequency response of the unit when tuned to 1.5MHz at the upper end of the MW band (a voltage gain of 50 is equivalent to a gain of 34dB).

From Fig.11.25, the measured bandwidth is approximately 80kHz.

The 9-12V DC supply is attached using a standard 5.5mm DC barrel connector, SK3. The front panel layout of the RF preamplifier is shown in Fig.11.26.

Table 11.2 shows the different ranges that can be obtained by changing the value of L1, while Table 11.3 provides some typical test voltages.

Conclusion

With that, we come to the end of this 11-part educational series on radio and radio communications.

Over nearly a year, we have covered topics including:

- Magnetic and electric fields
- Radio signal propagation
- Atmospheric effects on radio

Table 11.2: typical frequency ranges depending on the value of inductor L1

L1 value	Minimum	Maximum
220µH	470kHz	1.9MHz
100µH	690kHz	2.7MHz
33µH	1.2MHz	4.6MHz

Table 11.3: typical test voltages for 12V DC and 9V DC supplies

Part	Pin	@9V	@12V
TR1	d	8.1V	8.4V
	g2	5.0V*	5.1V*
	g1	0V	0V
	s	0.5V	0.5V
TR2	d	8.1V	8.4V
	g	4.0V	4.2V
	s	2.3V	2.4V

* varies with the setting of VR1

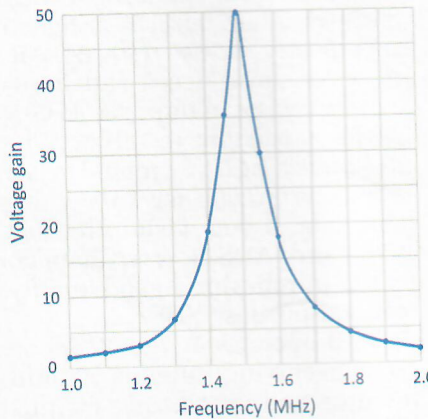


Fig.11.25: the frequency response of the high-gain RF preamplifier when tuned to 1.5MHz.

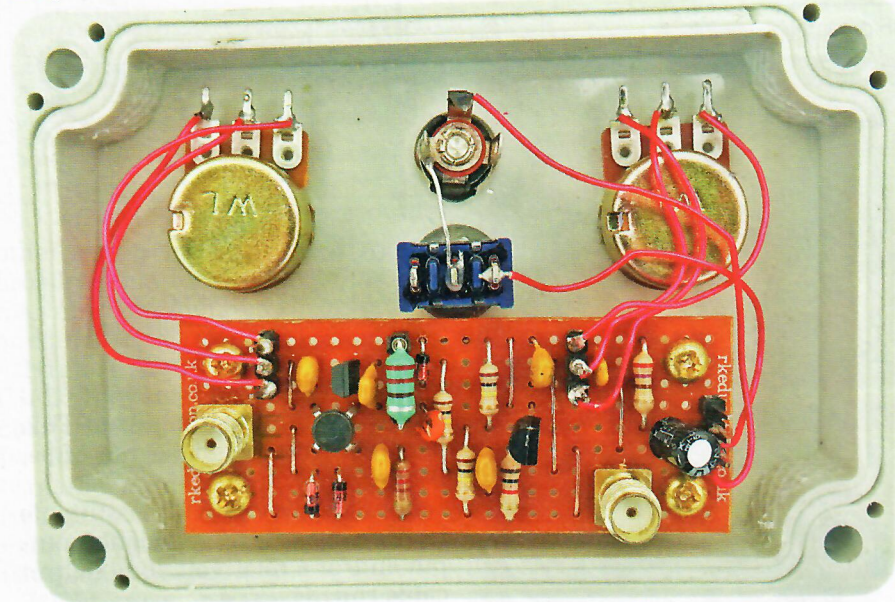


Fig.11.23: the internal wiring of the high-gain RF preamplifier.

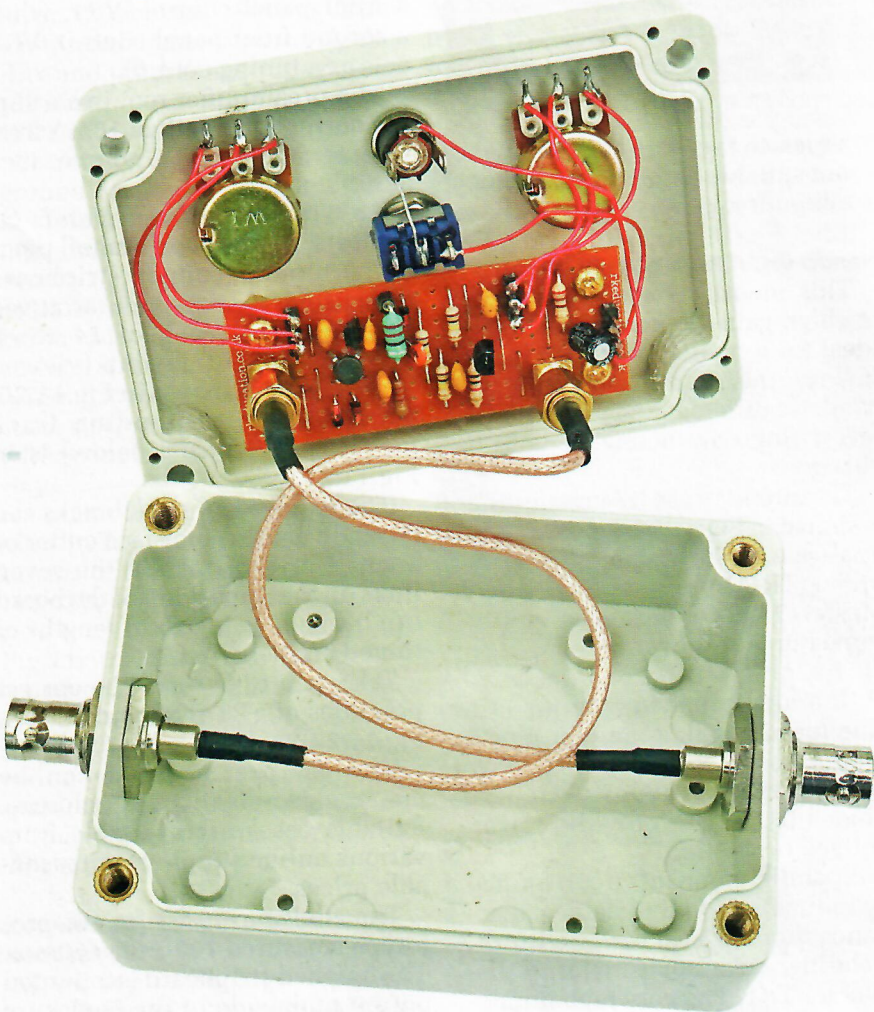


Fig.11.24: the wiring between the base and lid of the ABS enclosure.

- The ionosphere
- Building a modern 'crystal set'
- Tuned circuits
- Quartz crystals
- RF inductors and transformers
- RF semiconductors
- Modulation schemes
- Frequency conversion
- Software-defined radio
- Direct digital synthesis
- Antennas
- Baluns

- RF testing & measurement
- Spectrum analysis
- Vector network analysis
- Smith charts
- Frequency conversion
- Multipliers & mixers
- RF sources & oscillators
- Short-range RF devices
- Shortwave & amateur radio

We hope that you have found this series entertaining as well as enlightening!

PE

Parts List – High-gain RF preamplifier

- 2 10k Ω linear variable potentiometers (VR1 & VR2)
- 1 220 μ H axial lead inductor (L1) (see Table 11.3)
- 2 SMA female PCB mounting connectors (SK1 & SK2)
- 2 panel-mounting female BNC sockets with male SMA 15cm pigtail connectors (for bringing SK1 & SK2 to the outside of the case)
- 1 ABS plastic enclosure (minimum 100 × 68 × 50mm)
- 2 small plastic pointer knobs
- 4 brass or nylon M3 hexagonal 5mm stand-off spacers
- 4 short M3 panhead machined screws
- 4 short M3 countersunk head machined screws
- 1 5.5mm outer diameter male panel-mounting 9/12V DC barrel connector
- 1 miniature SPST toggle switch (S1)
- 3 3-way 0.1-inch/2.54mm pitch pin headers (P1, P2 & P3)

Semiconductors

- 1 BF961 dual-gate Mosfet (TR1)
- 1 2N7000 Mosfet (TR2)
- 2 1N4148 silicon switching diodes (D1 & D2)
- 1 1SV149 variable-capacitance diode (D3)

Capacitors

- 1 10 μ F 16V radial electrolytic (C8)
- 4 100nF 50V ceramic (C1, C2, C5 & C6)
- 2 10nF 50V ceramic (C3 & C7)
- 1 47pF 50V ceramic (C4)

Resistors (all fixed $\frac{1}{4}$ W axial, $\pm 5\%$ or better)

- 3 100k Ω (R2, R3 & R4)
- 1 1k Ω (R5)
- 1 220 Ω (R1)
- 1 150 Ω (R6)



Fig.11.26: the front panel layout of the high-gain RF preamplifier.

PoLabs

For more information or software download please visit 

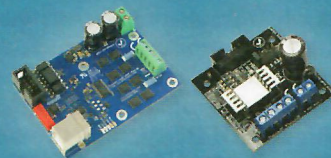
www.poscope.com/epe

PoKeys Connect, Control



- USB
- Ethernet
- Web server
- Modbus
- CNC (Mach3/4)
- IO
- PWM
- Encoders
- LCD
- Analog inputs
- Compact PLC

Stepper motor drivers Drive



- up to 256 microsteps
- 50 V / 6 A
- USB configuration
- Isolated
- up to 32 microsteps
- 30 V / 2.5 A

PoScope Mega1+ PoScope Mega50 Measure



- up to 50MS/s
- resolution up to 12bit
- Lowest power consumption
- Smallest and lightest
- 7 in 1: Oscilloscope, FFT, X/Y, Recorder, Logic Analyzer, Protocol decoder, Signal generator